

Assignment Day 4 - Functions

Sum Function

```
#include <iostream>
using namespace std;
int sum(int a,int b){ return a+b; }
int main(){ int a,b; cin>>a>>b; cout<<"Sum = "<<sum(a,b); return 0; }
```

Input: 10 20

Output: Sum = 30

Maximum of Two Numbers

```
#include <iostream>
using namespace std;
int maximum(int a,int b){ return (a>b)?a:b; }
int main(){ int a,b; cin>>a>>b; cout<<"Maximum = "<<maximum(a,b); return 0; }
```

Input: 15 9

Output: Maximum = 15

Prime Number Function

```
#include <iostream>
using namespace std;
bool isPrime(int n){
    if(n<=1) return false;
    for(int i=2;i<=n/2;i++) if(n%i==0) return false;
    return true;
}
int main(){ int n; cin>>n; cout<<(isPrime(n)?"Prime Number":"Not Prime Number"); }
```

Input: 13

Output: Prime Number

Factorial Function

```
#include <iostream>
using namespace std;
long long factorial(int n){ long long f=1; for(int i=1;i<=n;i++) f*=i; return f; }
int main(){ int n; cin>>n; cout<<"Factorial = "<<factorial(n); }
```

Input: 5

Output: Factorial = 120

Power Function

```
#include <iostream>
using namespace std;
long long power(int b,int e){ long long r=1; for(int i=1;i<=e;i++) r*=b; return r; }
int main(){ int b,e; cin>>b>>e; cout<<"Power = "<<power(b,e); }
```

Input: 2 5

Output: Power = 32

GCD Function

```
#include <iostream>
using namespace std;
int gcd(int a,int b){ while(b){ int t=b; b=a%b; a=t; } return a; }
int main(){ int a,b; cin>>a>>b; cout<<"GCD = "<<gcd(a,b); }
```

Input: 24 36

Output: GCD = 12